Tree Type Networks Revisited

Shared CNN features and small neural correction nodes

Aroma Indus
IIT Delhi

4 October 2026

Private code repository:
https://github.com/aromaindus049-byte/neural-tree-assignment

Overview

This study extends a parent-correction tree by replacing its classifiers with small neural networks. A compact CNN learns image features, which are subsequently frozen and shared across nodes. Children learn tasks derived from parent errors and feed signed corrections into the parent score. CIFAR-10 uses ten one-versus-rest roots and an airplane correction child; Cats vs Dogs uses a compact binary root with a downward correction child.

Final model	Parameters	Train	Validation	Test
CIFAR-10 correction forest	5,515,500	100.0000%	70.9600%	72.4000%
Compact Cats vs Dogs tree	37,891	99.9800%	78.6229%	77.2582%

Both models satisfy the near-perfect training-fit objective and exhibit large training-to-held-out gaps. The compact Cats vs Dogs tree improves training fit from 52 to four errors with an 85.8% parameter reduction relative to a larger root model. An audit of the residual errors identifies conflicting labels on identical preprocessed images, explaining a limit on exact interpolation.

Scope of evidence

A six-width CIFAR root-stage capacity sweep was performed, but double descent was not observed and exact interpolation was not reached by those root-only runs. Node visualizations use local input-gradient sensitivity rather than a complete theoretical receptive-field measurement. The reported results are recorded notebook outputs from local PyTorch MPS training; they were not independently retrained for this report.

1. Architecture and feature design

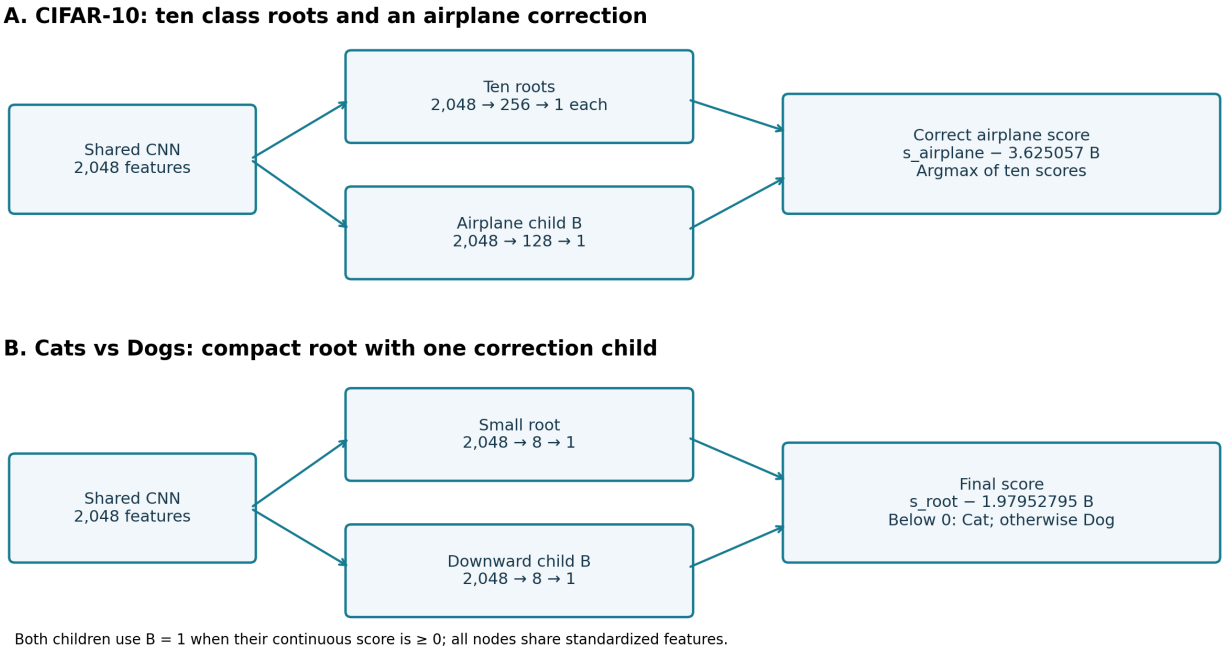


Figure 1. Final architectures. Shared features enter both parent and child networks. Child outputs modify parent scores; this is a correction tree, rather than a conventional left/right routing tree. CIFAR-10 is a collection of ten class trees, nine of which remain root-only.

The trunk is Conv2d(3,16,3,padding=1), ReLU, MaxPool2d(2), Conv2d(16,32,3,padding=1), ReLU, MaxPool2d(2), then flatten. A 32 x 32 RGB image becomes a 32 x 8 x 8 activation tensor and a 2,048-dimensional feature vector. The trunk contains 5,088 learned parameters. Separate trunks were trained for the two datasets; within each dataset the nodes share its trunk.

A node of hidden width h is Linear(2048,h), ReLU, Linear(h,1). It has 2,050h + 1 parameters. Features are standardized as z = (f - training mean) / max(training standard deviation, 10^{-6}). No validation or test features are used to estimate these statistics.

Parent errors define child tasks

Parent category	Meaning	Upward A target	Downward B target
C1	Predict 0; true 0	0	Ignore
C2	Predict 1; true 1	Ignore	0
C3	Predict 0; true 1	1	0
C4	Predict 1; true 0	0	1

The general score is s_corrected = s_parent + aA - bB + bias, with nonnegative correction strengths. Final models require only B, with zero additional bias. B is the hard threshold of a neural child score at zero. Correctly rejected negatives are excluded from B training, while genuine positives are included to discourage harmful corrections. The full training set is still used when selecting the correction strength.

2. Data, training, and implementation

Dataset	Training	Validation	Test	Preprocessing
CIFAR-10	45,000	5,000	10,000	ToTensor; native 32 x 32
Cats vs Dogs	19,998	2,498	2,502	RGB; resize 32 x 32; ToTensor

CIFAR-10 uses the official 50,000-image training pool with a random 45,000/5,000 split seeded at 42, and its official 10,000-image test set. Cats vs Dogs was downloaded from Microsoft: 12,500 JPEG files per class. Full RGB decoding excluded two unreadable files, leaving 12,499 per class. Each class was independently shuffled with NumPy seed 42 and split approximately 80/10/10. The saved split manifest records relative image paths. File paths do not overlap across splits; duplicate-content leakage across splits has not been audited.

Staged training

First train the CNN and a classifier jointly. For CIFAR-10, the preliminary classifier has a 64-neuron hidden layer and ten outputs, using cross-entropy. The corrected baseline run reached 66.24% training and 60.66% validation accuracy after 25 epochs. For Cats vs Dogs, a 128-neuron binary root and its CNN train for 20 epochs with BCEWithLogitsLoss and Adam at 0.001, reaching 86.65% training and 79.62% validation accuracy.

Next freeze the CNN, cache features in deterministic sample order, and train independent neural nodes on standardized features. CIFAR roots use weighted one-versus-rest BCE with each class positive weight equal to its negative/positive count, batches of 256, and 50 epochs. Adam starts at 0.001 and changes to 0.0001 after 25 epochs in the original root run. The largest raw class score determines the multiclass prediction; these weighted scores are not calibrated probabilities.

Cats vs Dogs feature roots use unweighted BCE, batches of 256, and 100 epochs, with learning rate reduced from 0.001 to 0.0001 at epoch 51. Child tasks use positive-class weighting to compensate for rare correction targets. The compact child restores its best checked balanced-task-error checkpoint, epoch 30. All correction strengths are chosen using training errors only; validation and test data are used for evaluation.

Practical implementation

Training ran in Jupyter on a MacBook Air using PyTorch MPS. num_workers=0 avoids notebook multiprocessing complications. Caching reduced the recorded 100-epoch Cats vs Dogs feature-root run to about 21 seconds, whereas image-based trunk/root training required roughly 30 seconds per epoch. Checkpoints store weights, widths, normalization, and correction strengths. These timings are single-machine observations, not benchmark claims.

The repository contains the exploratory notebook and README. Earlier cells include interactive reruns and debugging history; a clean restart-and-run-all pipeline is not verified. Model-initialization cells reset weights. Reproduction requires the dataset path to be updated and the intended execution order followed. Exact package versions were not recorded in the supplied notebook.

3. Final results and correction behavior

Model	Train errors / N	Validation errors / N	Test errors / N
CIFAR roots	1 / 45,000	1,451 / 5,000	2,761 / 10,000
CIFAR roots + B	0 / 45,000	1,452 / 5,000	2,760 / 10,000
Cats vs Dogs large root	2 / 19,998	505 / 2,498	Not evaluated
Cats vs Dogs small root	52 / 19,998	535 / 2,498	569 / 2,502
Cats vs Dogs compact tree	4 / 19,998	534 / 2,498	569 / 2,502

CIFAR-10: exact training interpolation

The ten original roots make one multiclass training error. The airplane root has 12 binary false positives and no missed positives; all other binary roots have zero threshold errors. Binary errors and multiclass errors differ because a false-positive class score need not be the largest score. A 128-neuron child targets the 12 false positives while protecting 4,512 real airplanes. At epoch 100 it flags all 12 false positives and 71 real airplanes.

A training-only grid of 201 nonnegative strengths selects $b = 3.625057$. Although some true airplanes activate the child, their remaining root scores preserve the correct decisions. The final model has zero training errors and removes all 12 binary airplane false positives. Validation loses one correct prediction; test gains one. These changes are too small to support a substantial generalization claim.

Cats vs Dogs: compact correction

A 128-neuron feature root reaches 99.99% training accuracy with two false dog predictions. Its 32-neuron child detects both but also activates on four genuine dogs. Training-error minimization chooses $b = 0$, so that attempted correction is inactive. The successful compact experiment starts from an 8-neuron root with 52 false dog predictions and no missed dogs. Its 8-neuron child detects all 52 and flags four genuine dogs.

Searching the nonnegative training-score breakpoints and intervening intervals selects $b = 1.97952795$. Subtraction removes all false dog predictions but creates four missed dogs. Root/child inference is float32; final Cats vs Dogs score subtraction is on CPU in float64, matching the evaluation code. The compact tree attains 99.98% training, 78.6229% validation, and 77.2582% test accuracy. Its training-validation gap is 21.36 percentage points; CIFAR has a 29.04-point gap.

Earlier recursive experiment

An exploratory CIFAR cat-versus-rest tree added upward/downward children and two grandchildren under B. Expansion changed training accuracy from 85.34% to 86.03%, and validation from 82.36% to 82.58%; validation balanced accuracy decreased from 79.23% to 78.21%. This separate experiment demonstrates recursion and the importance of class-balanced metrics, but is not the final ten-class model or the separate Cats vs Dogs dataset experiment.

4. Capacity, double descent, and size

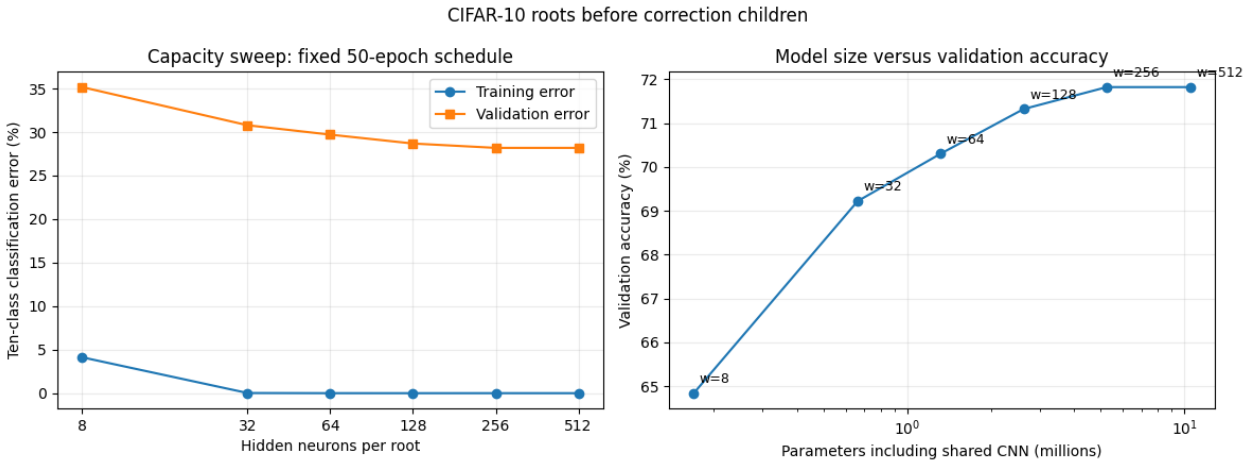


Figure 2. Recorded CIFAR root-stage sweep, before correction children. Near-zero plotted training errors are not necessarily exactly zero. Shared CNN features and a fixed 50-epoch schedule were used.

Root width	Parameters + CNN	Train errors	Validation accuracy
8	169,098	1,856	64.84%
32	661,098	10	69.22%
64	1,317,098	2	70.30%
128	2,629,098	1	71.32%
256	5,253,098	2	71.82%
512	10,501,098	1	71.82%

Validation error decreases from 35.16% to 28.18% and then plateaus. There is no peak followed by a second decline: capacity double descent was not observed. None of these root-only runs attained exact zero training error within the budget, so behavior after the exact interpolation threshold was not established. The sweep has one run per width, and optimization speed is confounded with capacity. The frozen trunk was trained on the same training set. No analogous Cats vs Dogs capacity sweep was performed.

Width 32 is the smallest tested CIFAR root stage exceeding 99.9% training accuracy: 99.9778%, with 87.4% fewer parameters than width 256. It has lower validation accuracy and was not expanded into an equivalent full correction forest. Width 512 doubles size relative to 256 without improving validation accuracy. The final CIFAR forest includes its child and strength: 5,088 + 5,248,010 + 262,401 + 1 = 5,515,500 parameters.

For Cats vs Dogs, the large root plus CNN uses 267,489 parameters. The small root uses 21,489; adding an 8-neuron child and one correction scalar gives 37,891. Thus the compact tree reduces parameters by 85.8% relative to the large root while retaining near-perfect fit. Normalization buffers, cached features, and optimizer states are excluded from learned-parameter counts. This is an exploratory size reduction, not a global minimum.

5. Insight: contradictory inputs limit interpolation

The two residual false dog predictions of the large Cats vs Dogs root have exactly matching CNN features and exactly identical RGB tensors after 32 x 32 preprocessing to examples labelled Dog. Maximum pixel difference is 0.0 in each checked pair.

Cat example	Matching Dog examples	Label counts	Minimum errors
Cat/8456.jpg	Dog/8736.jpg	1 Cat, 1 Dog	1
Cat/11565.jpg	Dog/2877.jpg; Dog/10797.jpg; Dog/10401.jpg	1 Cat, 3 Dogs	1

A deterministic classifier gives one prediction to one input. An identical-input group containing n_cat and n_dog labels therefore forces at least min(n_cat,n_dog) errors. These two groups impose at least two errors in the 19,998-image training set, giving accuracy no greater than (19,998 - 2)/19,998 = 99.989999%. The large root makes exactly two errors and therefore attains this upper bound for the present inputs.

This explains why adding neurons or correction children cannot fix the contradiction. The compact child marks the two Cat copies and their four Dog counterparts together. Its selected correction predicts Cat for these groups, producing four missed dogs instead of the two errors obtained by predicting Dog. It corrects many other small-root mistakes, but finishes two errors above the attainable minimum.

These findings apply to preprocessed tensors. The original full-resolution files were not compared, so original-file duplication or incorrect semantic labels cannot be asserted. No examples were removed or relabelled to improve reported accuracy. Grouping identical-content images before splitting, and auditing labels at original resolution, would be a better protocol for future experiments.

Conclusions and contributions

Shared learned features make neural correction trees practical on a laptop. Adaptive correction is useful for reducing training errors with small nodes, as shown by the 52-to-four error reduction in the compact tree. Perfect memorization does not establish useful generalization: final test gains are one image for CIFAR and zero net images for Cats vs Dogs. Data contradictions, rather than insufficient model capacity alone, can set an interpolation ceiling. These are experiment-specific interpretations; no claim of a new research architecture or statistical significance is made.

Future directions

Audit duplicates across all splits and group them before splitting; compare original-resolution contradictory examples. Repeat capacity sweeps with multiple seeds and longer convergence checks near interpolation, including active trees and Cats vs Dogs. Compare shared trunks with node-specific fine-tuning. Explore pruning, feature compression, and validation-guided adaptive child growth. Compare gradient maps with occlusion or integrated-gradient analyses and assess robustness across many examples.

6. Node receptive-field exploration: CIFAR-10

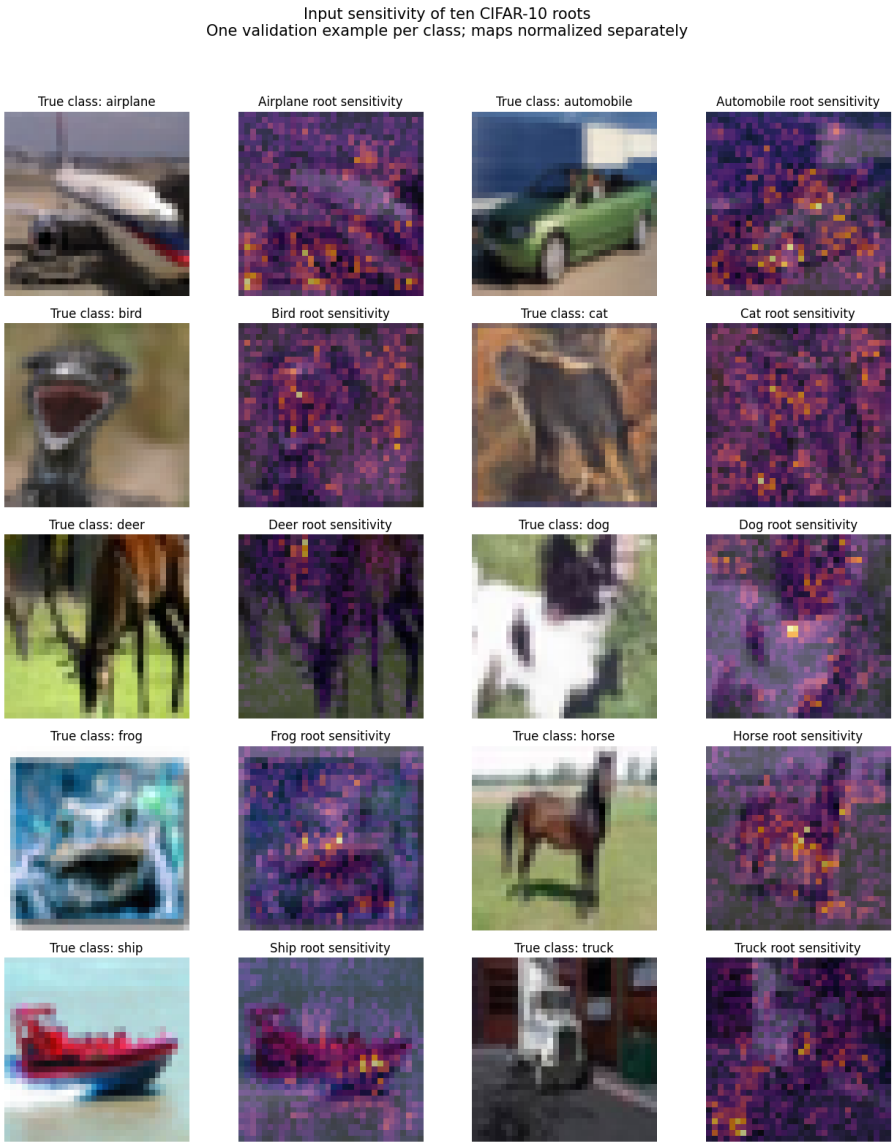


Figure 3. One validation example per class and its corresponding root sensitivity. Heatmaps are independently normalized. The airplane root in this figure is shown before its correction.

Sensitivity is abs(input gradient) maximized over RGB channels. Some strong responses overlap objects, such as the ship hull and horse body; others lie in background regions. Since both image and root vary between panels, this figure is illustrative and does not isolate root-specific differences on a common input.

7. Parent versus child sensitivity: CIFAR-10

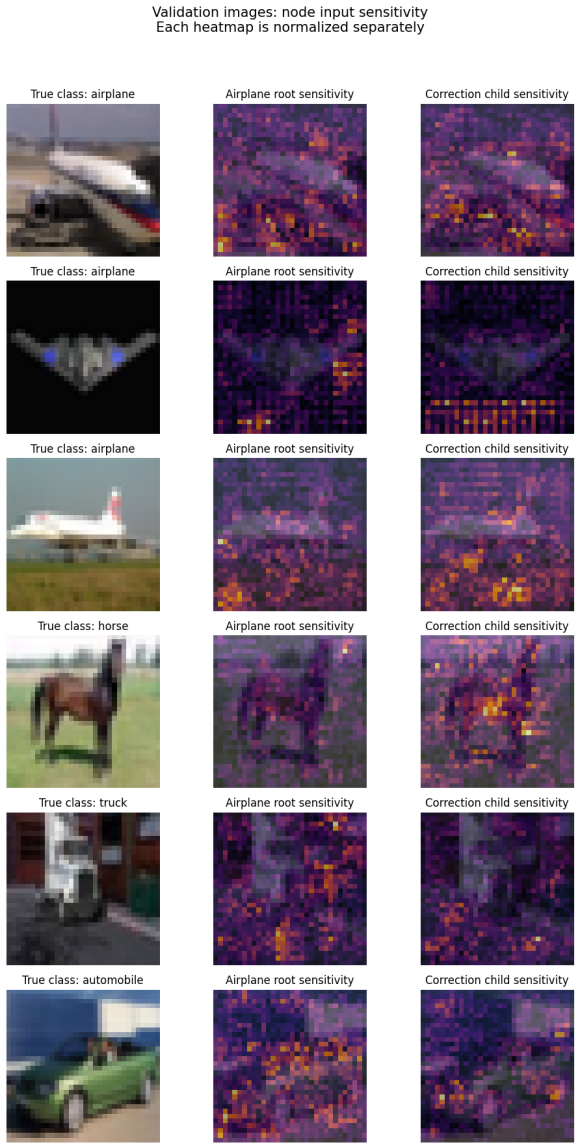


Figure 4. Airplane root and correction child on the same six validation images. Each node is differentiated through the shared frozen CNN using its continuous score, before any hard child threshold.

The child has a bright horizontal band below the second airplane, while the root responds near the right boundary. Both show background/ground sensitivity on the third airplane. On the horse, the child has a concentrated response around the body. These maps show local score sensitivity, not the sign of evidence or semantic attention. Independent normalization prevents quantitative comparison of brightness between nodes.

8. Parent versus child sensitivity: Cats vs Dogs

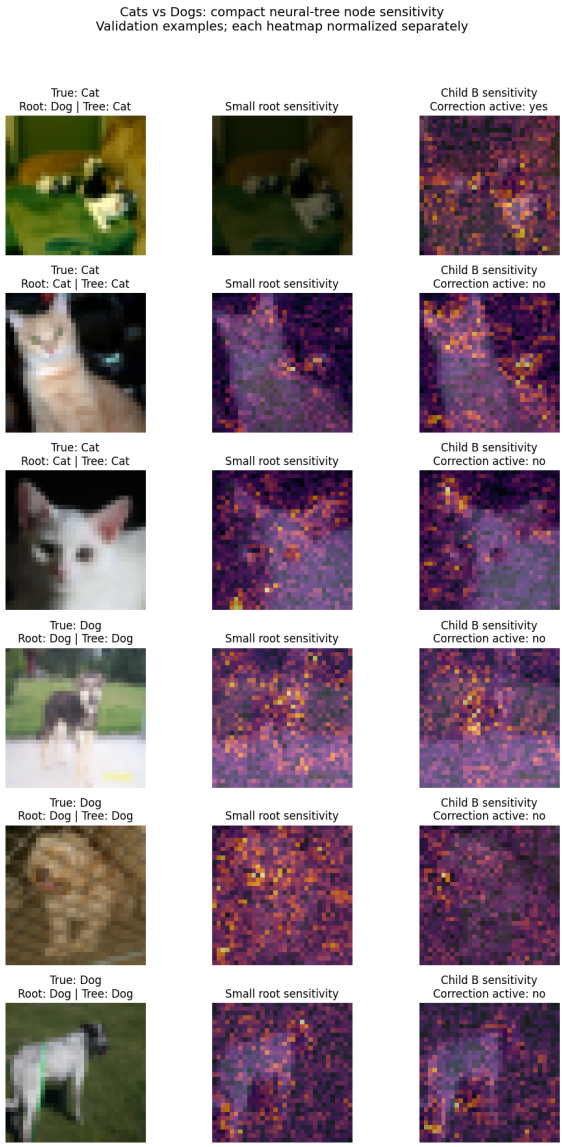


Figure 5. First three Cat and first three Dog validation examples. Labels record root/tree decisions and child activation; images were not selected by heatmap appearance.

In the first example, a Cat is predicted Dog by the root and corrected to Cat by child B. The other five predictions remain correct with the child inactive. Root and child maps differ and include animal and background regions. The first root map has no visible heatmap signal; local gradients may be zero in flat ReLU regions. This was not separately quantified. An inactive child can still have nonzero continuous-score sensitivity.

These maps explore empirical receptive-field behavior but do not measure the full theoretical field. Flattening followed by dense layers allows node scores to depend on the whole image, even though each final CNN location has a nominal 10 x 10 receptive field. Six examples cannot establish class-wide localization or explain the generalization gap causally.

Evidence and resources: recorded outputs in ASSIGNMENT TREE 3CNN.ipynb; CIFAR-10: https://www.cs.toronto.edu/~kriz/cifar.html ; Cats vs Dogs: https://www.microsoft.com/en-us/download/details.aspx?id=54 . Code and documentation: private repository linked on page 1. Reviewer access must be granted separately by the repository owner.